Gap-optimal conductances on ladder graphs

Theoretical documentation of the SDP solver in `gap-maximizing-on-ladders`

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Repository	`github.com/probabilist-jnieminen/gap-maximizing-on-ladders`
Code version documented	commit [TODO: hash] / tag [TODO: v0.x]
Document status	skeleton; sections marked TODO are unwritten

Abstract

[TODO: 3–5 sentences.] We document the mathematics behind a numerical solver for the problem of maximizing the algebraic connectivity λ_2 of an n-rung ladder graph over non-negative edge conductances with unit total budget. We state the problem, its semidefinite programming (SDP) formulation and dual, the optimality conditions used as numerical certificates, a symmetry reduction specific to ladders, and the design decisions and validation tests of the implementation.

Contents

1 Introduction and scope **2**

2 Preliminaries **2**
- 2.1 Weighted graphs and Laplacians . . . 2
- 2.2 Variational characterisation . . . 3
- 2.3 Ladder graphs . . . 3

3 The optimisation problem **4**

4 SDP formulation **4**

5 Duality and optimality conditions **5**

6 Symmetry reduction for ladders **5**

7 Physical interpretation: Heisenberg spin ladders **6**

8 Numerical method **7**
- 8.1 Modelling . . . 7
- 8.2 Solvers and tolerances . . . 7
- 8.3 Post-processing and certificates . . . 7
- 8.4 First-order alternative (optional) . . . 7

9 Validation **8**

10 Numerical observations **8**

11 Open questions and extensions **8**

A Notation and code **9**

B Design decision log **9**

1 Introduction and scope

Purpose. This document records (i) the mathematical results the solver relies on, with proofs or citations, (ii) the design decisions of the implementation and their reasons, and (iii) how the numerical output is validated.

Scope. Two-rail ladders $\mathcal{L}_n = P_n \,\square\, P_2$ with $n \geq 1$ rungs. Generalizations are listed in Section 11.

Map from sections to code.

Section	Content	Module(s)
Section 2	graph, incidence matrix, Laplacian	`ladder.py`, `laplacian.py`
Sections 4 and 5	SDP, dual, certificates	`solver.py` [TODO: confirm name]
Section 6	symmetry-reduced SDP	[TODO: module]
Section 8	solver pipeline, post-processing	[TODO: module]
Section 9	tests	`tests/`

Physical motivation. See Section 7: by Aldous' spectral gap theorem the problem is equivalent to choosing couplings of a ferromagnetic Heisenberg spin ladder that maximise its spectral gap.

2 Preliminaries

2.1 Weighted graphs and Laplacians

Let $G = (V, E)$ be a finite connected simple graph, $N = |V|$, $m = |E|$. Fix an arbitrary orientation of each edge $e = (i, j)$ and let $b_e = \boldsymbol{e}_i - \boldsymbol{e}_j \in \mathbb{R}^V$, $\boldsymbol{e}_i$ denoting the i-th standard basis vector (not to be mistaken for the edge e). The *incidence matrix* $B \in \mathbb{R}^{E \times V}$ has rows $b_e^\top$. Denote $\mathcal{S}_E := [0, 1]^E$, then for an arbitrary enumeration of E, let a *conductance vector* $w \in \mathcal{S}_E$ assign every edge in E a conductance between 0 and 1 based on its index in the enumeration. Define

$$L(w) \;=\; B^\top \operatorname{Diag}(w)\, B \;=\; \sum_{e \in E} w_e \, L_e, \qquad L_e := b_e b_e^\top. \tag{1}$$

For $e = (i, j)$, the image of said edge e under w will occasionally be denoted with $w_e := w_{ij}$.

Lemma 2.1 (Basic properties)**.** *For every $w \in \mathcal{S}_E$:*

(a) $L(w)$ coincides with the weighted graph Laplacian $D(w) - A(w)$ (weighted degree minus weighted adjacency) and does not depend on the chosen orientation;

(b) $x^\top L(w)\, x = \sum_{(i,j) \in E} w_{ij} (x_i - x_j)^2 \geq 0$;

(c) $L(w)\mathbf{1} = 0$ and $\operatorname{tr} L(w) = 2 \sum_{e \in E} w_e$;

(d) $w \mapsto L(w)$ is linear.

Proof. [TODO: Entrywise computation of $B^\top \operatorname{Diag}(w) B$; (b) from $x^\top L_e x = (b_e^\top x)^2$.] See also [3].

(a)

(b) Using (a),

(c)

(d) Using basic properties of matrices,

$$L(\alpha w_1 + \beta w_2) = B^\top \operatorname{Diag}(\alpha w_1 + \beta w_2)\, B = B^\top \left(\alpha \operatorname{Diag}(w_1) + \beta \operatorname{Diag}(w_2)\right) B$$
$$= \alpha B^\top \operatorname{Diag}(w_1) B + \beta B^\top \operatorname{Diag}(w_2) B = \alpha L(w_1) + \beta L(w_2).$$

□

Order the eigenvalues $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_N$. The *algebraic connectivity* is $\lambda_2(L(w))$ [7].

Lemma 2.2 (Connectivity)**.** *$\lambda_2(L(w)) > 0$ if and only if the graph $(V, \{e : w_e > 0\})$ is connected.*

Proof. [TODO: Kernel of $L(w)$ = functions constant on components, by Lemma 2.1(b).] [7] □

2.2 Variational characterisation

Lemma 2.3 (Courant–Fischer for λ_2)**.**

$$\lambda_2\big(L(w)\big) \;=\; \min_{\substack{x \perp \mathbf{1} \\ \|x\|=1}} x^\top L(w)\, x \;=\; \min_{\substack{x \perp \mathbf{1} \\ \|x\|=1}} \sum_{\{i,j\} \in E} w_{ij}(x_i - x_j)^2.$$

Proof. Courant–Fischer [14, Thm. 4.2.6] with the eigenvector $\mathbf{1}$ of $\lambda_1 = 0$ removed. [TODO: write out] □

Lemma 2.4 (Trace bound)**.** *If $\mathbf{1}^\top w = 1$ then $\lambda_2(L(w)) \le 2/(N-1)$, with equality iff $L(w)$ restricted to $\mathbf{1}^\perp$ is a multiple of the identity.*

Proof. λ_2 is at most the mean of $\lambda_2, \dots, \lambda_N$, which is $\operatorname{tr} L(w)/(N-1) = 2/(N-1)$ by Lemma 2.1(c). □

2.3 Ladder graphs

Definition 2.5 (Ladder)**.** For $n \ge 1$, $\mathcal{L}_n$ has vertices (i, s), $i \in \{1, \dots, n\}$, $s \in \{1, 2\}$, and edges

- *rails*: $\{(i,s), (i+1,s)\}$ for $1 \le i \le n-1$, conductance $\ell_i^{(s)}$;
- *rungs*: $\{(i,1), (i,2)\}$ for $1 \le i \le n$, conductance r_i.

Thus $N = 2n$, $m = 3n - 2$, and $\mathcal{L}_n \cong P_n \,\square\, P_2$.

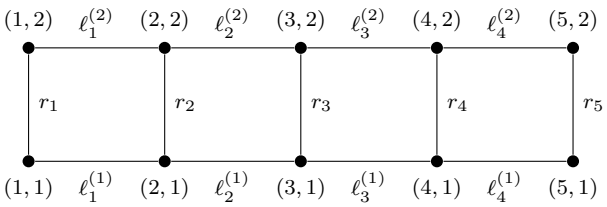


Figure 1: The ladder $\mathcal{L}_5$ with its edge labels.

Indexing convention. [TODO: State exactly how vertices (i, s) and edges map to integer indices and to rail/rung labels in `ladder.py`, so that formulas here can be checked against the code.]

Example 2.6 (Uniform conductances)**.** For $w \equiv 1/m$ the spectrum of $L(w)$ is

$$\Big\{ \tfrac{1}{3n-2}\big(2 - 2\cos\tfrac{\pi k}{n} + 2s\big) : 0 \le k \le n-1,\ s \in \{0,1\} \Big\},$$

since Laplacian spectra add under Cartesian products [3]. Hence $\lambda_2^{\mathrm{unif}} = \frac{2-2\cos(\pi/n)}{3n-2}$ for $n \ge 2$. This is the baseline against which optimised conductances are compared.

3 The optimisation problem

Let $\Delta_E = \{w \in \mathbb{R}^E_{\ge 0} : \mathbf{1}^\top w = 1\}$. The problem is

$$\lambda^\star(G) \;=\; \max_{w \in \Delta_E} \ \lambda_2\big(L(w)\big). \tag{P}$$

Proposition 3.1 (Concavity)**.** *$f(w) = \lambda_2(L(w))$ is concave on $\mathbb{R}^E_{\ge 0}$, so (P) is a convex optimisation problem.*

Proof. By Lemma 2.3, f is a pointwise minimum of the linear functions $w \mapsto \sum_e w_e (b_e^\top x)^2$. [TODO: tidy] □

Proposition 3.2 (Existence and bounds)**.** *The maximum in (P) is attained and $\lambda_2(L(\mathbf{1}/m)) \le \lambda^\star(G) \le 2/(N-1)$; in particular $\lambda^\star(G) > 0$.*

Proof. f is continuous and Δ_E is compact. Lower bound: feasibility of $\mathbf{1}/m$ and Lemma 2.2. Upper bound: Lemma 2.4. □

Remark 3.3 (Normalisation). Fiedler's *absolute algebraic connectivity* [8] uses the budget $\mathbf{1}^\top w = m$; by homogeneity it equals $m\,\lambda^\star(G)$.

Remark 3.4 (Nonsmoothness). f is differentiable at w only if $\lambda_2(L(w))$ is simple. At the optimum λ_2 is typically multiple, which is why the problem is solved as an SDP rather than by gradient ascent; see also Section 8.4.

4 SDP formulation

Let $P = I - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ (orthogonal projector onto $\mathbf{1}^\perp$) and let $Q \in \mathbb{R}^{N \times (N-1)}$ have orthonormal columns spanning $\mathbf{1}^\perp$, so $QQ^\top = P$ and $Q^\top Q = I_{N-1}$.

Lemma 4.1. *For $w \ge 0$ and $t \in \mathbb{R}$ the following are equivalent: (i) $\lambda_2(L(w)) \ge t$; (ii) $L(w) - tP \succeq 0$; (iii) $Q^\top L(w)\, Q \succeq t\, I_{N-1}$.*

Proof. [TODO: $L(w)$ and P commute and share the eigenvector $\mathbf{1}$ (eigenvalue 0 for both); on $\mathbf{1}^\perp$, P acts as the identity.] □

Consequently (P) equals each of the SDPs

$$\begin{aligned} &\text{maximise} && t \\ &\text{subject to} && L(w) - tP \succeq 0, \quad w \ge 0, \quad \mathbf{1}^\top w = 1, \end{aligned} \tag{SDP$_P$}$$

$$\begin{aligned} &\text{maximise} && t \\ &\text{subject to} && Q^\top L(w)\, Q - t I_{N-1} \succeq 0, \quad w \ge 0, \quad \mathbf{1}^\top w = 1, \end{aligned} \tag{SDP$_Q$}$$

both of which are linear in (w, t) by Lemma 2.1(d) [17, 1].

Remark 4.2 (Design choice). [TODO: Which formulation the code uses and why (matrix size N vs. $N-1$, sparsity of $L(w)$ vs. density of Q, conditioning, ease of extracting the dual). How Q is built, e.g. `scipy.linalg.null_space`.]

5 Duality and optimality conditions

The Lagrange dual of (SDP_P) is

$$\begin{aligned} &\text{minimise} && \mu \\ &\text{subject to} && \langle L_e,\, X\rangle \le \mu \quad (e \in E), \qquad \langle P,\, X\rangle = 1, \qquad X \succeq 0. \end{aligned} \tag{D}$$

[TODO: Derive via the Lagrangian $t + \langle X,\, L(w) - tP\rangle + \mu(1 - \mathbf{1}^\top w)$.]

Lemma 5.1 (Gap identity). *For (w,t) feasible for (SDP_P) and (X,μ) feasible for (D),*

$$\mu - t \;=\; \langle L(w) - tP,\, X\rangle \;+\; \sum_{e\in E} w_e\big(\mu - \langle L_e,\, X\rangle\big) \;\ge\; 0.$$

Proof. Expand using $\mathbf{1}^\top w = 1$ and $\langle P,\, X\rangle = 1$; both terms are nonnegative. [TODO: write out] □

Theorem 5.2 (Strong duality). *Both (SDP_P) and (D) attain their optima, and the optimal values coincide and equal $\lambda^\star(G)$.*

Proof. [TODO: Slater: $w = \mathbf{1}/m$ with $t < \lambda_2(L(\mathbf{1}/m))$ is strictly feasible for the primal; $X = P/(N-1)$ with large μ for the dual.] [17, 2] □

Corollary 5.3 (Optimality conditions). *Feasible $(w^\star, t^\star)$ and $(X^\star, \mu^\star)$ are optimal iff $t^\star = \mu^\star$, iff*

(i) $\big(L(w^\star) - t^\star P\big)X^\star = 0$, *and*

(ii) $w_e^\star > 0 \;\Rightarrow\; \langle L_e,\, X^\star\rangle = \mu^\star$.

Proposition 5.4 (Embedding form of the dual). *Without loss of generality $X\mathbf{1} = 0$ in (D). Writing $X = VV^\top$ with rows $v_i \in \mathbb{R}^d$,*

$$\frac{1}{\lambda^\star(G)} \;=\; \max\Big\{\textstyle\sum_{i\in V}\|v_i\|^2 \;:\; \sum_{i\in V} v_i = 0,\quad \|v_i - v_j\| \le 1 \;\;\forall\{i,j\}\in E\Big\}.$$

Proof. [TODO: Replace X by PXP (objective and constraints unchanged since $L_eP = L_e$); then $\langle L_e,\, X\rangle = \|v_i - v_j\|^2$ and $\langle P,\, X\rangle = \sum_i \|v_i\|^2$; rescale.] This is the "maximum variance" problem of [16, 10]. □

Remark 5.5 (Interpretation). By Corollary 5.3(i), every coordinate of an optimal embedding is a $\lambda^\star$-eigenvector of $L(w^\star)$, so $\operatorname{rank} X^\star \le \operatorname{mult}\big(\lambda_2(L(w^\star))\big)$. By (ii), every edge carrying positive conductance has length exactly 1 in the scaled embedding. [TODO: Check [11] for a bound on the embedding dimension in terms of tree-width; $\mathcal{L}_n$ has tree-width 2.]

Remark 5.6 (Numerical certificate). Any $w \in \Delta_E$ gives a lower bound $\lambda_2(L(w)) \le \lambda^\star$ and any $X \succeq 0$ with $X\mathbf{1} = 0$, $\operatorname{tr} X = 1$ gives an upper bound $\lambda^\star \le \max_e \langle L_e,\, X\rangle$. Solver output is repaired to exact feasibility (Section 8.3) to obtain a certified bracket.

6 Symmetry reduction for ladders

Lemma 6.1 (Automorphisms). *For $n \ge 3$, $\operatorname{Aut}(\mathcal{L}_n) = \langle \sigma, \rho\rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, where $\sigma(i,s) = (i, 3-s)$ (rail swap) and $\rho(i,s) = (n+1-i, s)$ (reversal). For $n = 2$, $\mathcal{L}_2 \cong C_4$ and the group is dihedral of order 8.*

Proof. [TODO: Corners are the only vertices of degree 2 …] □

Proposition 6.2 (Symmetric optima exist)**.** *Let $\Gamma \leq \mathrm{Aut}(G)$ and $w^\star$ be optimal for (P). Then the average $\bar{w} = \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma} \gamma w^\star$ is optimal and Γ-invariant.*

Proof. $L(\gamma w)$ is permutation-similar to $L(w)$, so $f(\gamma w^\star) = \lambda^\star$; Δ_E is Γ-invariant; conclude by concavity (Proposition 3.1). See [9] for symmetry reduction of SDPs in general. □

Hence we may restrict to σ-invariant conductances $\ell_i := \ell_i^{(1)} = \ell_i^{(2)}$ $(1 \leq i \leq n-1)$ and r_i $(1 \leq i \leq n)$, with budget $2\,\mathbf{1}^\top \ell + \mathbf{1}^\top r = 1$. Let A_ℓ be the Laplacian of the path P_n with conductances ℓ, and $R = \mathrm{Diag}(r)$.

Proposition 6.3 (Sector decomposition)**.** *Order vertices as $(1,1),\dots,(n,1),(1,2),\dots,(n,2)$ and let $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} I_n & I_n \\ I_n & -I_n \end{smallmatrix}\right)$. For σ-invariant w,*

$$L(w) = \begin{pmatrix} A_\ell + R & -R \\ -R & A_\ell + R \end{pmatrix}, \qquad U^\top L(w)\, U = \begin{pmatrix} A_\ell & 0 \\ 0 & A_\ell + 2R \end{pmatrix}.$$

Proof. [TODO: Direct block multiplication.] □

Corollary 6.4. *For σ-invariant w,*

$$\lambda_2\big(L(w)\big) = \min\big\{\lambda_2(A_\ell),\ \lambda_1(A_\ell + 2R)\big\},$$

with the convention $\lambda_2(A_\ell) = +\infty$ when $n = 1$. The first term is the symmetric sector *(rails move together; rungs play no role); the second is the* antisymmetric sector*, a discrete Schrödinger operator on P_n with potential $2r$.*

The reduced SDP has two LMIs of sizes $n-1$ and n instead of one of size $2n-1$:

$$\begin{aligned} \text{maximise} \quad & t \\ \text{subject to} \quad & Q_n^\top A_\ell\, Q_n \succeq t\, I_{n-1}, \qquad A_\ell + 2\,\mathrm{Diag}(r) \succeq t\, I_n, \\ & \ell \in \mathbb{R}^{n-1}_{\geq 0}, \quad r \in \mathbb{R}^n_{\geq 0}, \quad 2\,\mathbf{1}^\top \ell + \mathbf{1}^\top r = 1. \end{aligned} \tag{SDP$_{\mathrm{red}}$}$$

Example 6.5 (Closed forms for $n = 1, 2$)**.** $n = 1$: only the rung, $r_1 = 1$, so $\lambda^\star(\mathcal{L}_1) = 2$. $n = 2$: $\lambda_2(A_\ell) = 2\ell_1$ and $\lambda_1(A_\ell + 2R) = 2r$ (with $r_1 = r_2 = r$ by ρ-symmetry), under $\ell_1 + r = \frac{1}{2}$; hence $\ell_1 = r = \frac{1}{4}$, i.e. uniform conductances, and $\lambda^\star(\mathcal{L}_2) = \frac{1}{2}$.

Remark 6.6 (Further reduction). Imposing ρ-invariance ($\ell_i = \ell_{n-i}$, $r_i = r_{n+1-i}$) splits each sector further into even and odd parts under reversal. [TODO: Work out; decide whether worth implementing.]

Remark 6.7 (Heuristic). Budget moved from rails to rungs raises the antisymmetric sector but lowers the symmetric one, so at the optimum one expects both sectors to be active, i.e. $\mathrm{mult}(\lambda_2) \geq 2$. [TODO: Prove or check numerically.]

7 Physical interpretation: Heisenberg spin ladders

For couplings $J \in \mathbb{R}^E_{\geq 0}$ consider the spin-$\frac{1}{2}$ ferromagnetic Heisenberg Hamiltonian on $(\mathbb{C}^2)^{\otimes V}$,

$$H_J = \sum_{\{i,j\}\in E} J_{ij}\Big(\tfrac{1}{4} - \boldsymbol{S}_i \cdot \boldsymbol{S}_j\Big) = \tfrac{1}{2}\sum_{\{i,j\}\in E} J_{ij}\,(1 - T_{ij}),$$

where T_{ij} swaps the tensor factors i and j.

Theorem 7.1 (Aldous' spectral gap conjecture [4])**.** *The spectral gap of the interchange process with rates J equals that of the single random walk, $\lambda_2(L(J))$. Consequently the spectral gap of H_J is $\frac{1}{2}\lambda_2(L(J))$ in the normalisation above.*

[TODO: Align normalisation with the thesis preprint; explain the representation-theoretic step (the gap is attained in the standard representation, which occurs in $(\mathbb{C}^2)^{\otimes N}$).] Thus (P) with $G = \mathcal{L}_n$ is exactly the problem of gap-optimal couplings for a Heisenberg spin ladder under a total-coupling budget.

8 Numerical method

8.1 Modelling

The SDP is written in CVXPY [5], with $L(w)$ assembled as $B^\top \mathrm{Diag}(w)B$ from the NetworkX [13] incidence matrix. [TODO: Short code excerpt or pseudo-code of `solve_ladder(n)`; which variables and constraints; how the dual X is read from the LMI constraint, including CVXPY's sign and scaling conventions.]

8.2 Solvers and tolerances

Solver	Method	Notes
SCS [15]	first-order (ADMM)	scales to larger n; lower accuracy
Clarabel [12]	interior point	open source; higher accuracy
MOSEK	interior point	commercial, free academic licence

[TODO: Which solver is the default and why; tolerance values used.]

Symbol	Meaning	Value
$\varepsilon_{\text{feas}}$	feasibility tolerance	[TODO:]
$\varepsilon_{\text{supp}}$	threshold for $w_e > 0$ (support)	[TODO:]
$\varepsilon_{\text{mult}}$	eigenvalue clustering for $\mathrm{mult}(\lambda_2)$	[TODO:]
ε_{gap}	accepted certified gap $\hat{\mu} - \hat{\lambda}$	[TODO:]

8.3 Post-processing and certificates

1. *Primal repair:* $w \leftarrow \max(w, 0)$, $w \leftarrow w/\mathbf{1}^\top w$, so $w \in \Delta_E$ exactly.
2. *Lower bound:* $\hat{\lambda} = \lambda_2(L(w))$ computed with `numpy.linalg.eigvalsh`, a dense symmetric eigensolver.
3. *Dual repair:* clip negative eigenvalues of X, replace $X \leftarrow PXP/\operatorname{tr}(PXP)$; upper bound $\hat{\mu} = \max_e \langle L_e, X\rangle$ (Remark 5.6).
4. *Diagnostics:* certified gap $\hat{\mu} - \hat{\lambda}$; support $\{e : w_e > \varepsilon_{\text{supp}}\}$; $\mathrm{mult}(\lambda_2)$; residuals of Corollary 5.3(i)–(ii); optimal embedding from the eigendecomposition of X.

Algorithm 1 Solve and certify (P) for $\mathcal{L}_n$

Require: $n \geq 1$; solver $\mathcal{S}$; tolerances

build $\mathcal{L}_n$, B, Q ▷ `ladder.py`, `laplacian.py`
solve (SDP_Q) or ($\mathrm{SDP}_{\mathrm{red}}$) with $\mathcal{S}$; read w, t and dual X
repair w and X; compute $\hat{\lambda}$ and $\hat{\mu}$
if $\hat{\mu} - \hat{\lambda} > \varepsilon_{\text{gap}}$ **then**
    flag result as uncertified ▷ [TODO: policy]
end if
compute support, $\mathrm{mult}(\lambda_2)$, KKT residuals, embedding
return w, $[\hat{\lambda}, \hat{\mu}]$, diagnostics

8.4 First-order alternative (optional)

If x is a unit eigenvector of $L(w)$ for λ_2 with $x \perp \mathbf{1}$, then $g_e = (b_e^\top x)^2$ is a supergradient of f at w: indeed $f(w') \leq x^\top L(w')x = g^\top w' = f(w) + g^\top(w' - w)$. Projected supergradient ascent on Δ_E (projection as in [6]) gives a simple solver whose iterates can be animated by the visualiser. [TODO: Step sizes, convergence guarantee, comparison with CVXPY.]

9 Validation

Test	Checks	File
$L(w) = B^\top \mathrm{Diag}(w)B$ vs. NetworkX Laplacian	Lemma 2.1	[TODO:]
Uniform spectrum	Example 2.6	[TODO:]
$n = 1$: $\lambda^\star = 2$; $n = 2$: $\lambda^\star = \frac{1}{2}$, uniform w	Example 6.5	[TODO:]
$\lambda_2^{\mathrm{unif}} \le \hat{\lambda} \le \lambda^\star \le \hat{\mu} \le 2/(2n-1)$	Proposition 3.2 and Remark 5.6	[TODO:]
certified gap below $\varepsilon_{\mathrm{gap}}$	Lemma 5.1 and Theorem 5.2	[TODO:]
symmetrised solution is optimal	Proposition 6.2	[TODO:]
full and reduced SDP agree	Corollary 6.4	[TODO:]
SCS and Clarabel agree	solver cross-check	[TODO:]

10 Numerical observations

[TODO: Fill after Phase 2. Decide what may appear publicly before the thesis results are cleared.]

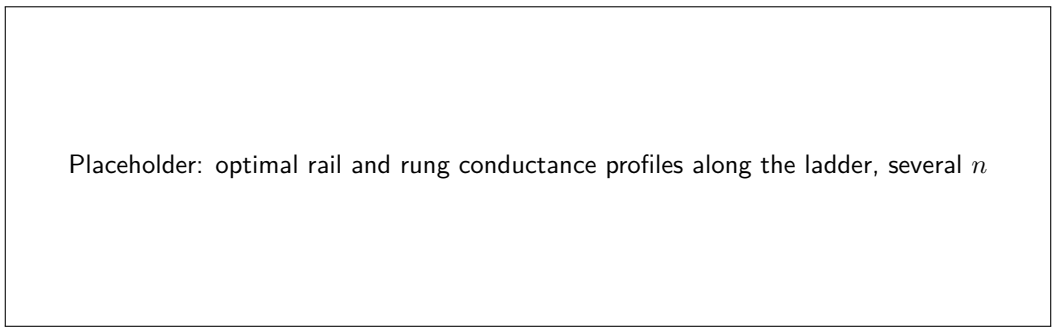


Figure 2: [TODO: caption]

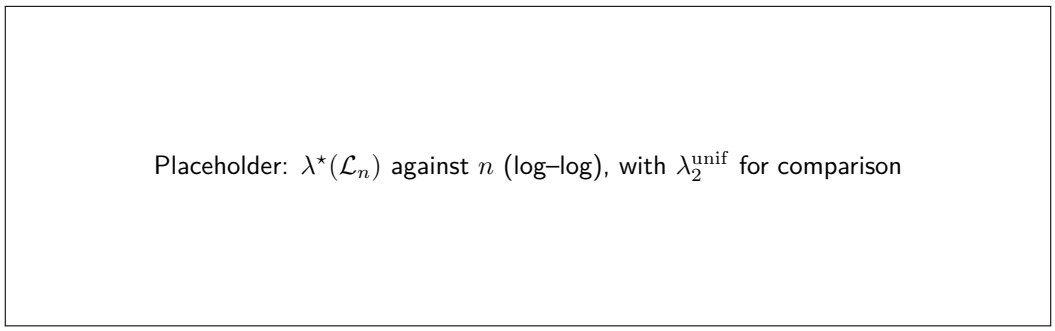


Figure 3: [TODO: caption]

Conjecture 10.1 (Monotonicity)**.** *[TODO: State precisely.]*

11 Open questions and extensions

- Asymptotics of $\lambda^\star(\mathcal{L}_n)$ as $n \to \infty$ and of the optimal profiles. [TODO:]
- Ladders with k rails, $P_n \,\square\, P_k$; circular ladders $C_n \,\square\, P_2$.
- Other budgets, e.g. $\sum_e d_e^2 w_e \le 1$ as in [16].
- Quantum side: [TODO: links to the preprint and the QC project].

A Notation and code

Symbol	Meaning	Code
n	number of rungs	`n` in `solve_ladder(n)`
$\mathcal{L}_n$	ladder graph	[TODO: function in `ladder.py`]
B	incidence matrix ($m \times N$)	[TODO:]
$L(w)$	weighted Laplacian	[TODO: function in `laplacian.py`]
w; ℓ, r	conductances; rail and rung parts	[TODO:]
Q, P	basis of / projector onto $\mathbf{1}^\perp$	[TODO:]
X	dual matrix / Gram matrix of embedding	[TODO:]
$\hat{\lambda}$, $\hat{\mu}$	certified lower and upper bounds	[TODO:]

B Design decision log

Decision	Alternatives	Reason
NetworkX for graph and incidence matrix; own rail/rung labelling	custom graph class	established library keeps code short and readable; NetworkX has no notion of rails and rungs, hence the small custom layer
$L(w) = B^\top \operatorname{Diag}(w) B$	assemble $D - A$ entry-wise	affine in w, so usable directly in CVXPY; matches Lemma 2.1
CVXPY as modelling layer	call a solver API directly	near-mathematical syntax; solver can be swapped with one argument
(SDP_P) vs. (SDP_Q)	[TODO:]	[TODO:]
default solver	[TODO:]	[TODO:]
flat package layout (no `src/`)	`src/` layout	matches the repository as created [TODO: expand]

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